

INTRODUCTION TO FLUID MECHANICS

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The ejector shown in the figure is used to discharge a container filled with a perfect liquid of density ρ and constant specific heat c into the atmosphere, also at pressure p_a . A high-velocity jet of the same liquid, with velocity U_j and cross section A_j , is injected along the axis of the discharge pipe, whose cross section is A . The jet entrains fluid as it develops, producing an underpressure at the injection section and drawing the fluid from the container. Mixing of the jet with the surrounding liquid is completed downstream, so that at the outlet the liquid discharges as a uniform jet with velocity U_o . For a given the geometry (i.e., given values of A_j and A), we want to compute the characteristics of the flow in terms of the injection velocity U_j . In the computation, assume that the effect of gravity is negligible. Address in order the following questions:

1. Assuming that the acceleration of the fluid upstream from the injector is steady and inviscid, write down the equation that relates the ambient pressure p_a with the velocity and pressure at the entrance of the pipe, U_e and p_e .
2. Apply the continuity and momentum balance equations to the control volume indicated in the figure. Simplify the result by assuming that the effect of friction on the wall surface is negligibly small.
3. Use the previous equations to show that the volumetric discharge flow rate is given by

$$Q = U_e(A - A_j) = \frac{U_j(A - A_j)}{1 + A/\sqrt{2A_j(A - A_j)}}$$

and that the velocity at the outlet satisfies

$$\frac{U_o}{U_j} = \frac{A_j + \sqrt{2A_j(A - A_j)}}{A + \sqrt{2A_j(A - A_j)}}$$

4. Assuming that the values of the temperatures T_e and T_j at the entrance section are equal to the ambient value T_a , obtain the temperature T_o at the outlet section when the pipe walls are adiabatic.

1) $p_a = p_e + \frac{1}{2} \rho U_e^2$ (1)

2) $-\rho U_e(A - A_j) - \rho U_j A_j + \rho U_o A = 0$ (2)

$-\rho U_e^2(A - A_j) - \rho U_j^2 A_j + \rho U_o^2 A = (p_e - p_a)A - \int_{\text{wall}} \vec{\tau} \cdot \vec{n} dS = -\frac{1}{2} \rho U_e^2 A$ (3)

3) $U_o A = U_e(A - A_j) + U_j A_j \rightarrow \text{From (2)}$

$U_o^2 A = U_e^2 \left(\frac{A}{2} - A_j \right) + U_j^2 A_j \rightarrow \text{From (3)}$

$U_o^2 A^2 = U_e^2 (A^2 - 2AA_j + A_j^2) + U_j^2 A_j^2 + 2A_j(A - A_j)U_e U_j = A \left(\frac{A}{2} - A_j \right) U_e^2 + AA_j U_j^2$

4) USING THE ENERGY EQUATION

$\rho U_o A \left(cT_o + \frac{U_o^2}{2} \right) - \rho U_j A_j \left(cT_j + \frac{U_j^2}{2} \right) - \rho U_e(A - A_j) \left(cT_e + \frac{U_e^2}{2} \right) =$

$= p_e U_e(A - A_j) + p_e U_j A_j - p_a U_o A \stackrel{(2)}{=} (p_e - p_a) U_o A = -\frac{1}{2} \rho U_e^2 p_a U_o A$

IF $T_j = T_e = T_a$

$\frac{c(T_o - T_a)}{U_j^2} = \frac{U_j A_j}{U_o A} \left[1 - \left(\frac{U_e}{U_j} \right)^2 \right] - \left(\frac{U_o}{U_j} \right)^2$

$\frac{U_e}{U_j} = \frac{1}{1 + A/\sqrt{2A_j(A - A_j)}}$

$\frac{U_o}{U_j} = \frac{A_j + \sqrt{2A_j(A - A_j)}}{A + \sqrt{2A_j(A - A_j)}}$

$Q = \frac{U_j(A - A_j)}{1 + A/\sqrt{2A_j(A - A_j)}}$